How the ranking works and what the data cannot tell you

1 Only three tests have a CPCB bathing limit: dissolved oxygen (5 mg/l or more), BOD (3 mg/l or less) and pH (6.5 to 8.5). Only these are scored.

2 A reading within its limit scores 0. A reading that breaks it scores 50, plus up to 50 more for how far past the limit it is: oxygen at 0 scores 100.

3 Durga readings count 1.5×. 2024 and 2025 count 1.25× and 1.5×. Missing readings count nothing either way.

4 Risk score = 50% oxygen + 30% BOD + 20% pH, from 0 to 100.

5 Act now: score 55 or more, or any oxygen reading below 2 mg/l. High: 30 or more. Moderate: below 30. There is no "clean" tier: every ghat broke the BOD limit.

6 Action: aerate + test if oxygen fell below 2 and Durga 2025 is missing; test first if Durga 2025 is missing; aerate if immersion pulled oxygen below 5 or it stayed low a week later; extra cleaning + re-test if BOD or pH broke the limit.

7 Robustness: the ranking is recomputed under 24 scoring choices (3 CPCB classes × Durga weight × year weight × parameter weights).

8 Checks: 18 build-time checks reproduce the problem statement figures, and 1,467 printed values were found on their cited pages in the MPPCB PDFs.

Does the ranking depend on our choices?

1. Bairagarh (Sant Hirdaram) kund: #1 in all 24 scoring choices
2. Malikhedi kund: Rank 2–4 across 24 scoring choices
3. Shahpura lake ghat: Rank 2–5 across 24 scoring choices
4. Rani Kamlapati ghat / Kamla Park kund: Rank 2–5 across 24 scoring choices
5. Khatlapura ghat: Rank 4–7 across 24 scoring choices
6. Hathaikheda dam / kund: Rank 4–6 across 24 scoring choices
7. Prempura ghat: Rank 5–7 across 24 scoring choices

What this data can't tell you

- No health claims. The reports test no bacteria (coliform), so the tool compares readings with CPCB limits and says nothing about illness.
- Blank, _, * and ** mean not sampled. They show as missing and never count as clean. BDL means below detection: very low, not zero.
- Bairagarh and Malikhedi have no Durga 2025 table and were tested only during Ganesh 2025 immersion. Their ranks rest on 13 of 18 tests.
- "Rani Kamlapati ghat / Kamla Park kund" may be two places: Ganesh tables say Kamla Park kund, Durga tables say Lower Lake ghat.
- Units are assumed. The MPPCB reports print none; the standard unit for each test is used.
- Only four metals were tested (chromium, lead, zinc, copper). CPCB sets no bathing limit for them, so they are shown but not scored.
- Each festival has three samples. A single sample a day can miss what happens between samples.
- CPCB Class B (outdoor bathing) is used because people stand in the water during immersion. The ranking was also checked under Classes C and D.
- The NGT has banned immersion in the Upper and Lower Lakes and BMC is building new ghats (Free Press Journal, 26 Feb 2026). Lake ghats here may not be used this year; their readings still describe the lake water.
- The statewide file is less checked. Unreadable sites, conflicting duplicates and 14 impossible oxygen values were removed before comparing.
- One CSV/PDF difference was found: Bairagarh, Ganesh 2025, after-immersion pH is ** in the CSV and * in the PDF. Both mean not sampled, so nothing changes.
- The score is a planning aid for where to act first. It is not a legal finding of compliance.

18 data checks passed at build

✓ Bhopal readings: 1524
✓ Unique reading IDs: 1524
✓ Ghats: 7
✓ Parameters: 13
✓ Dissolved-oxygen readings with a number: 116
✓ Dissolved oxygen below 5 mg/l: 42
✓ BOD readings with a number: 112
✓ BOD above 3 mg/l: 100
✓ Zero-oxygen readings: 4
✓ Zero-oxygen readings all at Bairagarh: true
✓ Zero-oxygen years: 2023,2025
✓ Below-detection (BDL) readings: 174
✓ Missing readings (not sampled, *, **): 61
✓ CPCB limits: 19
✓ CPCB Class B limits: 4
✓ Class B dissolved oxygen minimum: 5
✓ Class B BOD maximum: 3
✓ Class B pH range: 6.5-8.5

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